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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC603/20EC601 – ANTENNA AND MICROWAVE SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the construction of strip lines with a neat diagram and necessary equations.	10	L*2	1	1
	b) Explain construction of micro-strip line and losses associated with micro strip line with necessary equations.	10	L2	1	1
2.	a) Derive the condition for impedance matching in transmission lines with all necessary equations of voltage and current.	10	L2	1	1
	b) Explain the S parameters for a two port network.	10	L2	2	1
3.	a) A load impedance of $Z_R=60-j80\Omega$ is required to be matched to a 50Ω coaxial line, by using a short circuited stub of length "l" located at a distance 'd' from the load. The wavelength of operation is 1 metre. Using Smith chart, find 'd' and 'l'.	10	L3	1	1
	b) Explain E-plane Tee construction. Derive the S matrix for an E-plane Tee junction.	10	L2	2	1
Unit – II					
4.	a) Explain the parameters related to the antenna with necessary equations (i)Radiation intensity (ii)Beam area (iii)Directivity (iv)Gain	10	L2	3	1
	b) Derive the relation between directivity and beam area.	10	L2	3	1
5.	a) Derive array factor for 'n' isotropic point sources of equal amplitude and spacing.	10	L2	3	1
	b) Derive and plot the radiation pattern for an array of two isotropic sources of equal amplitude and opposite phase.	10	L2	4	1
6.	a) Derive an expression for two isotropic sources of equal amplitude and spacing.	10	L2	3	1
	b) Derive the equation for scalar potential for short dipole.	10	L2	4	1
Unit – III					
7.	a) Explain the working of 6 element-Yagi-uda antenna with necessary equations.	10	L2	5	1
	b) Explain the working of slot antennas with neat sketch.	10	L2	5	1
8.	a) Explain the working of log-periodic antenna with necessary equations.	10	L2	5	1
	b) Explain the working of reflector antenna with necessary equations.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
